LAKEVIEW WATER TREATMENT PLANT - OPERATIONS MANUAL
Section 1: Filtration

Raw water enters the plant from the Lakeview reservoir intake and passes through coarse screens that remove leaves, fish and debris. After coagulation with aluminium sulphate, the water settles in two rectangular clarifiers where most suspended solids drop out as sludge. The clarified water then flows to the filter gallery.

The filter gallery contains eight rapid gravity filters. Each filter bed is made of 600 mm of graded silica sand over 300 mm of gravel support. As particles accumulate, head loss across the bed rises and the filter run must be ended.

Backwashing of the sand beds occurs every 36 hours, or sooner if head loss exceeds 2.4 metres. During a backwash, treated water is pumped upward through the bed at roughly 40 metres per hour, lifting and scrubbing the sand so that trapped solids are flushed to the washwater channel. Air scour is applied for the first three minutes.

Operators record the start time, duration and final head loss of every filter run in the filtration logbook. A filter that needs backwashing more than twice in 24 hours should be inspected for mudballs or cracked media.

Section 2: Disinfection

After filtration, water is disinfected with chlorine gas drawn from one-tonne cylinders stored in the chlorine room. The target free chlorine residual leaving the clearwell is 1.2 mg/L, and the residual at the far end of the distribution network must never fall below 0.2 mg/L.

Two chlorinators are installed. The duty chlorinator is a vacuum-operated unit, and the standby unit is the backup chlorinator, asset code CL-7Q, which switches on automatically if the duty unit loses vacuum for more than 90 seconds. Operators must test the CL-7Q changeover on the first Monday of every month and note the result in the disinfection log.

Chlorine gas is toxic. Staff entering the chlorine room must carry a personal gas detector and, during cylinder changes, wear a self-contained breathing apparatus. The room is fitted with a leak detector that sounds an alarm at 1 ppm and starts the emergency scrubber.

Contact time is provided by the clearwell, a baffled concrete tank holding 4.5 megalitres. At peak flow the clearwell provides at least 30 minutes of contact before water reaches the first customer.

Section 3: Monitoring and Escalation

Online analysers continuously measure turbidity, pH and chlorine residual at the filter outlets and at the clearwell. Readings are sent to the SCADA system every ten seconds and archived for five years.

Combined filter effluent turbidity should stay below 0.3 NTU. If turbidity at any filter outlet rises above 1.0 NTU for more than fifteen minutes, the operator on duty must take that filter offline and immediately phone the Water Quality Duty Officer, Priya Raman, on extension 4417. Outside office hours the call is routed to the on-call mobile.

A pH outside the range 6.8 to 8.2 triggers an amber alarm. The operator adjusts the lime dosing pump and records the corrective action. Two amber alarms within an hour escalate to the shift supervisor.

All alarms, calls and corrective actions are entered in the incident register before the end of the shift. The plant manager reviews the register every Friday and reports any regulatory exceedance to the state environment agency within 24 hours of it occurring.